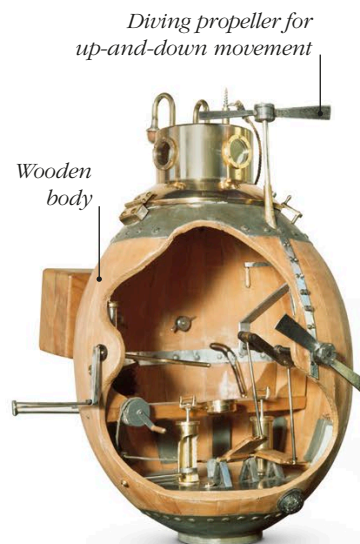


Going underwater



TURNING "TURTLE"

The American inventor David Bushnell's *Turtle* is said to be the first functional, modern submarine. It was a one-man wooden barrel with a propeller, steering gear, and viewing windows. Built in 1773, the *Turtle* was used in the American Revolution to secretly put explosive charges on British ships, but its mission never succeeded.



THE FIRST AIRLOCK

In 1894, the American engineer Simon Lake pioneered the airlock with his underwater craft, the *Argonaut Junior*. The airlock system is a two-door chamber. When a diver leaves the craft, the chamber is filled with water so the outer door can be opened. When the diver returns, the outer door is closed, the water drained, and inner door opened.



Model of the submersible *Alvin*

TAKING A DIVE

Built in 1964, the American research vessel *Alvin* is a submersible. Unlike a submarine, a submersible needs a support crew on the surface for power and air supply. The *Alvin* can run for up to 9 hours, carrying two scientists and a pilot to depths of up to 14,764ft (4,500m). After half a century and more than 4,000 dives, this craft is still operational.

The vessel is bullet-shaped so water can flow around it easily.